

BodyCheck — LotusHacks 2026 Master Plan

Event: LotusHacks × HackHarvard × GenAI Fund

Date: March 20–22, 2026 | VNG Campus, Ho Chi Minh City

Duration: 30 hours (MVP scope)

Team: 4 people

What We're Building

BodyCheck is an AI-powered medical triage web app. Users tap a part of their body that hurts, answer 3–5 follow-up questions, and receive an instant AI-generated diagnosis with severity rating, recommended actions, and real medical references — all readable aloud in Vietnamese or English.

LotusHacks Track Alignment

Track	Sponsor	How BodyCheck Qualifies
Social & Mobility	TASCO	Healthcare accessibility for underserved Vietnamese communities
Technology & Consumer	AWS	Mobile-first consumer health app with AI
Enterprise	TinyFish	B2B triage tool for clinics and insurance companies
EdTech	ETEST	Medical education: learn anatomy by exploring symptoms

Primary track: Social & Mobility. **Pitch all four** in the deck.

Team Files (read yours)

Person	Role	File
Person 1	UI Lead — HTML/CSS + Body Map	PERSON1_UI.md
Person 2	Form + State + Voice Input	PERSON2_FORM.md
Person 3	AI Core — LLM + TTS	PERSON3_AI.md
Person 4	AI Extras + Deploy + Pitch	PERSON4_DEPLOY.md

Tech Stack (Final)

Frontend

- **HTML5 + CSS3** — no framework, plain files
- **Vanilla JavaScript** — no build step needed for MVP
- **Lovable.dev** — AI-generate the base HTML shell in minutes (free, sponsor)
- **react-body-highlighter** — pre-mapped SVG body map with clickable regions

AI Services (all free)

Service	Use	Get Key At
OpenRouter	Claude + Qwen LLM calls	openrouter.ai → Sign up → Keys
fal.ai	Body region image generation	fal.ai → Dashboard → API Keys
ElevenLabs	Voice input (STT) + readback (TTS)	elevenlabs.io → Profile → API Key

Repository Structure

```
bodycheck/
├── index.html           ← Person 1 owns
├── styles/
│   ├── main.css        ← Person 1 owns
│   └── diagnosis.css    ← Person 1 owns
├── components/
│   ├── body-map.js      ← Person 1 owns
│   ├── symptom-form.js  ← Person 2 owns
│   ├── voice-input.js   ← Person 2 owns
│   └── diagnosis-card.js ← Person 1 owns (UI), Person 3 wires data
├── api/
│   ├── openrouter.js    ← Person 3 owns
│   ├── elevenlabs.js    ← Person 3 owns
│   ├── fal.js           ← Person 4 owns
│   └── exa.js           ← Person 4 owns
├── state/
│   └── app-state.js      ← Person 2 owns
├── data/
│   └── body-regions.js   ← Person 1 owns
├── .env.example         ← Person 3 creates
└── README.md            ← Person 4 writes
```

30-Hour Timeline (All Four)

Hour	0-1	[ALL]	Kickoff – roles, repo, API keys, .env setup
Hour	1-3	[P1]	Lovable.dev scaffold → HTML shell
Hour	1-3	[P3]	OpenRouter account + first test call
Hour	3-7	[P1]	Body map SVG integration
Hour	3-7	[P2]	Symptom form HTML + CSS
Hour	7-9	[P2]	App state wiring (JS object)
Hour	7-11	[P3]	LLM prompt + JSON parse + error handling
Hour	9-11	[P1]	Diagnosis card UI
Hour	11-13	[P3]	ElevenLabs TTS integration
Hour	11-14	[P4]	fal.ai image generation
Hour	14-17	[P4]	Exa.ai medical references
Hour	17-19	[P2]	ElevenLabs STT voice input
Hour	19-21	[P2]	Vietnamese language mode (Qwen swap)
Hour	21-24	[P1]	Mobile CSS polish
Hour	24-26	[ALL]	Integration testing + bug fixes
Hour	26-27	[P4]	Replit deploy + public URL
Hour	27-29	[P4]	Demo video recording + pitch deck
Hour	29-30	[ALL]	Rehearse demo, final check

API Keys Needed (Hour 0)

Person 3 collects all keys and shares via team chat BEFORE anyone writes API code.

```
OPENROUTER_KEY=sk-or-...      # openrouter.ai  
FAL_KEY=...                   # fal.ai  
ELEVENLABS_KEY=...           # elevenlabs.io  
EXA_KEY=...                   # exa.ai
```

Store in `.env` file. Never commit to GitHub (add to `.gitignore`).

Medical Disclaimer (Required Everywhere)

"BodyCheck is an AI tool for general information only and is not a substitute for professional medical advice, diagnosis, or treatment. If you are experiencing a medical emergency, call 115 immediately."

This text MUST appear: in every AI system prompt, on the diagnosis card, and in the app footer.

How to Win

1. **Demo works live** — judges watch the full flow: tap → speak → diagnose → hear result
2. **6+ AI tools visible** — call out fal, ElevenLabs, Exa, OpenRouter in the pitch
3. **Vietnamese language** — shows local impact, huge in Vietnam context
4. **Hit all 4 tracks** — spend 30 seconds per track in the pitch
5. **Sponsor tools = sponsor prizes** — double the prize pool eligibility